



ELEMENTARY MATH (CALCULUS)



WORKING WITH REAL-VALUED FUNCTIONS

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LEARNING OBJECTIVES

By the end of this lesson, you should be able to:

- Define a real-valued function with examples.
- Evaluate the domain, range, inverse and composite of a function.
- Interpret graphical representations of functions.
- Identify real world applications of different types of functions and explain how they model various phenomena.





INTRODUCTION

- Real-valued functions serve as the building blocks of mathematics. Functions offer a precise way to represent relationships between quantities.
- They allow mathematicians to describe how one variable depends on another, enabling the modeling and analysis of complex phenomena.





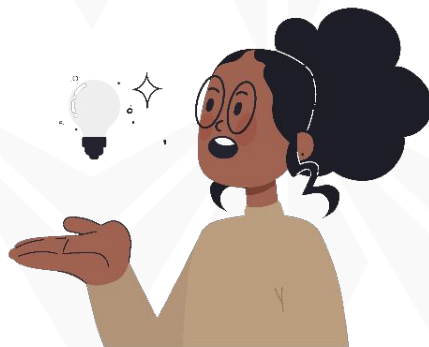
IMAGINE THIS

Imagine you're at an ice cream parlor, and you're presented with a menu of ice cream flavors.

Each flavor corresponds to a specific combination of ingredients and taste.

If you choose vanilla, you expect the taste of vanilla, and if you choose chocolate, you anticipate a chocolatey experience.

This concept of pairing choices (inputs) with outcomes (outputs) is at the heart of real-valued functions in mathematics.





WHAT ARE REAL-VALUED FUNCTIONS?

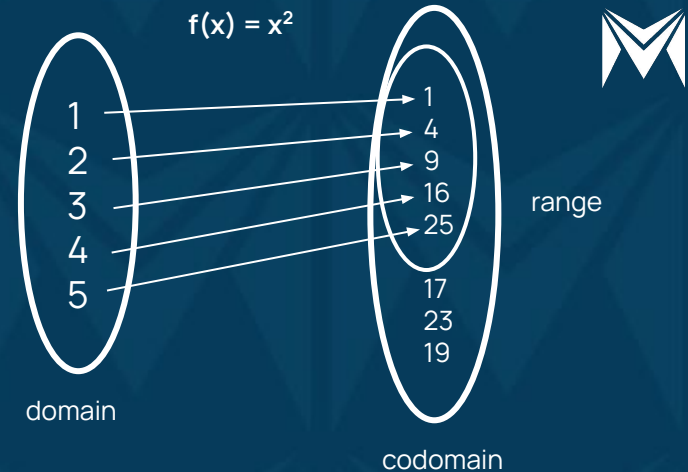
It is a mathematical rule or relationship that assigns a single, real number (the union of both rational and irrational numbers) to each element in a specific set.

The function f from X to Y is denoted by

$$f: X \rightarrow Y$$

Where, X is the “domain” and Y is the “co-domain”.

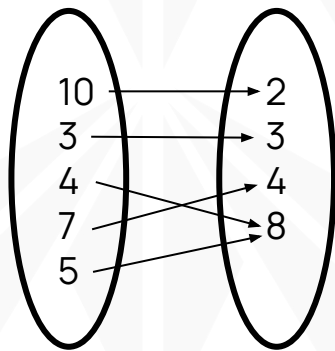
Therefore, a function whose “range” is a set of real numbers is called a REAL-VALUED FUNCTION.





A FUNCTION OR NOT?

Given two non-empty sets, X and Y



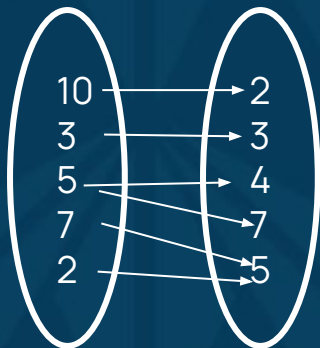
This is a function because there is **exactly one** output for every input.

RANGE = $\{2,3,4,8\}$



A FUNCTION OR NOT?

Given two non-empty sets, A and B



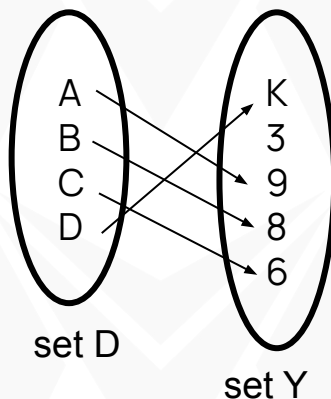
This is NOT a function because there is **more than one** output for a given input.

RANGE = $\{2,3,4,5,7\}$



IS THIS A REAL-VALUED FUNCTION?

Given two non-empty sets, D and Y



This is a function but **not** a REAL-VALUED FUNCTION because k is not a real number.

DOMAIN = $\{A, B, C, D\}$

RANGE = $\{6, 8, 9, k\}$,

CO-DOMAIN = $\{3, 6, 8, 9, k\}$



QUESTION

Is function the same as relation?

functions
(sets of ordered pairs where each input has only one output)

relations
(all sets of ordered pairs)





EXAMPLES OF REAL-VALUED FUNCTIONS

1. LINEAR FUNCTION:

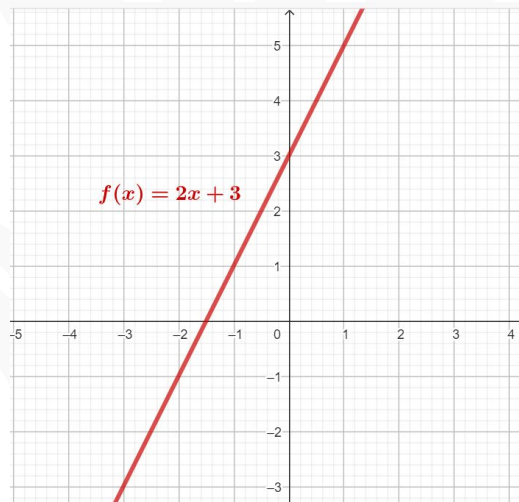
$$\text{Given } f(x) = 2x + 3$$

In this linear function, for each input x , we multiply it by 2 and then add 3 to get the output $f(x)$.

If you plug in $x = 2$,
you'll get

$$f(2) = 2(2) + 3 = 4 + 3 = 7.$$

So, for an input of 2, the output is 7.





EXAMPLES OF REAL-VALUED FUNCTIONS

2. QUADRATIC FUNCTION:

$$\text{Given } f(x) = x^2 - 4$$

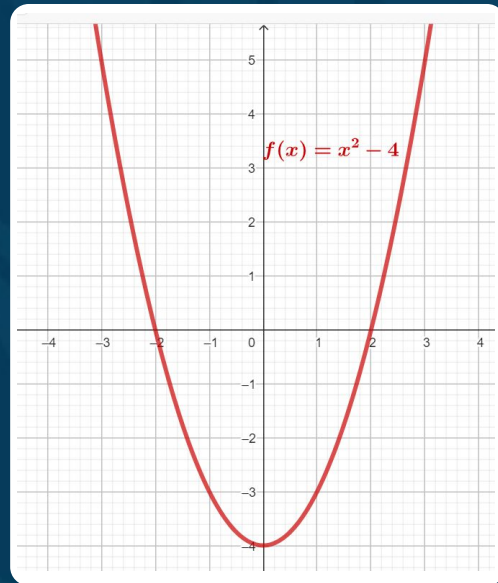
This quadratic function takes the square of the input x and subtracts 4.

if you input $x = 3$,

you'll calculate

$$f(3) = (3)^2 - 4 = 9 - 4 = 5$$

so, for an input of 3, the output is 5.





HOW TO FIND THE DOMAIN OF A FUNCTION

Example 1: Find the domain of a function

$$f(x) = x^2 + 3x + 2$$

Solution:

We know that the domain of a function is the set of all possible allowable input values i.e. *x values*.

Therefore,

$$D = (-\infty, \infty)$$



HOW TO FIND THE DOMAIN OF A FUNCTION

Example 2: What is the domain of $f(x) = \frac{3x+1}{x^2+5}$

Solution:

For $f(x)$ to be a defined function, the denominator can not be zero i.e.

$$x^2 + 5 \neq 0$$

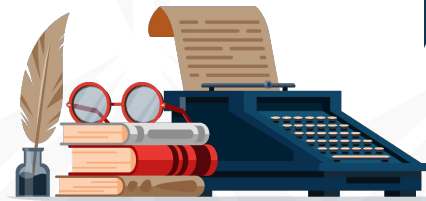
$$x^2 \neq -5$$

NOTE: the square of both positive and negative numbers will always give a positive number. Thus,

$$D = (-\infty, \infty)$$



HOW TO FIND THE DOMAIN OF A FUNCTION



Example 3: Given $f(x) = \sqrt{x+2}$

Solution: Take all the expression under the root sign, i.e.

$$x + 2 \geq 0$$

$$x \geq -2$$

Because, we can not have a negative number under the root sign for a real-valued function.

Thus, $D = [-2, \infty)$



HOW TO FIND THE RANGE OF A FUNCTION

Example 1: Find the range of $f(x) = 2x - 9$

Solution:

We know that the range is the set of all possible output values (typically represented by y or $f(x)$).

$$y = 2x - 9$$

$$2x = y + 9$$

$$x = \frac{y + 9}{2}$$

Thus, $R = (-\infty, \infty)$



HOW TO FIND THE RANGE OF A FUNCTION



Example 2: What is the range of $g(x) = \frac{3}{x+8}$

Solution: Let $g(x) = y$

$$y = \frac{3}{x+8}$$

$$y(x+8) = 3$$

$$xy + 8y = 3$$

$$xy = 3 - 8y$$

$$x = \frac{3 - 8y}{y}$$

Since $y \neq 0$

The range, $y \in \mathbb{R} - \{0\}$



HOW TO FIND THE RANGE OF A FUNCTION

Example 3: Find the range of $f(x) = 2 - |x - 3|$

Solution:

Let

$$y = 2 - |x - 3|$$

For an absolute function, if $|x - 3| \geq 0$

Then, $-|x - 3| \leq 0$

Add 2 to both sides

$$2 - |x - 3| \leq 0 + 2$$

$$2 - |x - 3| \leq 2$$

$$y \leq 2$$

Thus, $R = (-\infty, 2]$



TASK

What is the domain and range of $h(x) = \sqrt{x+2} - 8$



HOW TO FIND THE INVERSE OF A FUNCTION

Example 1: Find the inverse of $f(x) = 3x - 7$

Solution:

Let

$$y = 3x - 7$$

Switch x with y

$$x = 3y - 7$$

Then, isolate the y variable

$$\begin{aligned} 3y &= x + 7 \\ y &= \frac{x + 7}{3} \end{aligned}$$

Thus,

$$f^{-1} = \frac{x + 7}{3}$$



HOW TO SOLVE A COMPOSITE FUNCTION

Example 1: Given $f(x) = 5x - 4$ and $g(x) = x^2$.

What is $f \circ g$?

Solution:

$$\begin{aligned}f \circ g &= f[g(x)] \\f[x^2] &= 5(x^2) - 4 = 5x^2 - 4\end{aligned}$$

Example 2: Given $f(x) = 5x - 4$ and $g(x) = x^2$.

What is $g \circ f$?

Solution:

$$\begin{aligned}g \circ f &= g[f(x)] \\g[5x - 4] &= (5x - 4)^2 \\&= (5x - 4)(5x - 4) \\&= 25x^2 - 20x - 20x + 16 \\&= 25x^2 - 40x + 16\end{aligned}$$



GRAPHICAL REPRESENTATION

Each point on the graph corresponds to an input-output pair: $(x, f(x))$.

For example, consider the graph of the function $f(x) = x^2$.

As we increase x , the value of $f(x)$ increases exponentially. The graph shows a parabolic curve opening upward.

- Graphs are incredibly powerful because they allow us to see patterns, trends, and relationships in a visual way. They're not just about lines and curves; they're about understanding how things change.



QUESTION

Graph $f(x) = x + 2$, when $x = 0, 1, 2$

Solution:

Using a table, when $x = 0, y = 2$

$x = 1, y = 1,$

$x = 2, y = 0$





CONCLUSION

In conclusion, domain and range define where our function lives, function notation gives our function a name, and graphical representation helps us visualize the behavior of the function.

These concepts are essential tools for understanding and working with functions in mathematics and various real-world applications.



ELEMENTARY MATH II (CALCULUS)



SUMMARY



SUMMARY

- A function is a mathematical relation that assigns exactly one output value to each input value.
- Functions take an input (called the domain) and produce an output (called the range).
- Functions are typically represented as x , where x is the input, and $f(x)$ is the output.
- Functions can be graphically represented, with the x-axis representing input and the y-axis representing output.



FURTHER READING

Larson, R., & Edwards, B. H. (2017). *Calculus*. Cengage Learning.

Stewart, J. (2015). *Essential Calculus*. Cengage Learning.



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THANK
YOU